

"Kestrel catalyst efficiency below threshold P0420 from degraded converter"

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Area: bulletins

1 Condition

The Kestrel may set DTC P0420 (Catalyst System Efficiency Below Threshold, Bank 1) with an illuminated MIL and otherwise normal driveability. The code indicates the downstream oxygen sensor signal is tracking the upstream signal too closely, showing the catalyst is no longer storing and releasing oxygen effectively. Confirm P0420 is current and not a stored history code from a prior repair. Begin with the Kestrel Emissions Service Manual :: Emissions System Diagnosis to rule out exhaust leaks and sensor faults before condemning the converter.

2 Cause

The primary cause of P0420 is thermal aging or contamination of the catalytic converter, part number CAT-5500, which reduces its oxygen storage capacity below the threshold the PCM monitors. A secondary contributor is a lazy or biased downstream oxygen sensor, part number O2D-5120, whose slow switching can mimic a degraded catalyst. Verify O2D-5120 response before replacing CAT-5500, since a faulty sensor will set the same code without any converter failure. The Kestrel Emissions Service Manual :: Catalyst Efficiency Faults and the Kestrel Emissions Service Manual :: Upstream and Downstream Oxygen Sensors cover the discrimination tests.

3 Repair Procedure

When the catalyst is confirmed degraded, replace converter CAT-5500 following PROC-CAT, and replace the downstream sensor O2D-5120 if its switching response is out of specification. Inspect upstream gaskets and flanges for leaks that would corrupt the efficiency calculation, then clear P0420 and complete a drive cycle to verify the monitor passes. Torque the converter fasteners to specification and avoid contaminating the new O2D-5120 threads. Full instructions are in the Kestrel Emissions Service Manual :: Catalytic Converter Replacement Procedure and the converter identification appears in the Kestrel Emissions Service Manual :: Catalytic Converter (Bank 1 and Bank 2) section. See the parent Kestrel Emissions Service Manual for monitor drive-cycle requirements.